



Pack the Prism Teaching Bundle

Grade 5 • North Carolina • Mathematics

NC.5.MD.5 • Connect layers of cubes with volume.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Connect layers of cubes with volume. This bundle practices a focused part of NC.5.MD.5; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Count cubes in a base layer, then multiply by the number of layers. State volume in cubic units.

Example 1: A storage prism for the library team measures 4 cm by 3 cm by 3 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

One layer has $4 \times 3 = 12$ cubes. 3 layers give $12 \times 3 = 36 \text{ cm}^3$.

Example 2: A storage prism for the art club measures 7 cm by 4 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

One layer has $7 \times 4 = 28$ cubes. 2 layers give $28 \times 2 = 56 \text{ cm}^3$.

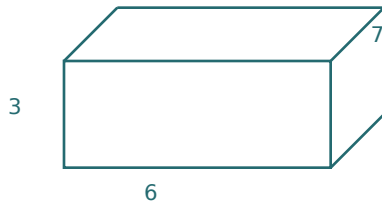
Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1–4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

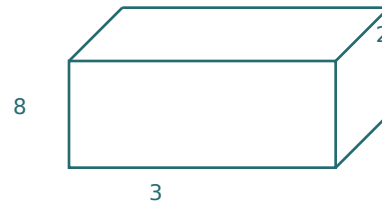
A storage prism for the art club measures 6 cm by 3 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 02

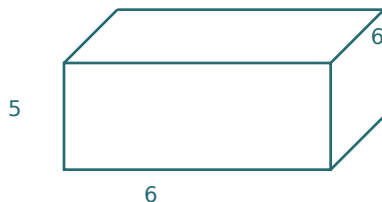
A storage prism for the science club measures 3 cm by 8 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 03

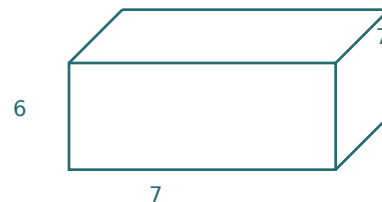
A storage prism for the community center measures 6 cm by 5 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 04

A storage prism for the reading group measures 7 cm by 6 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



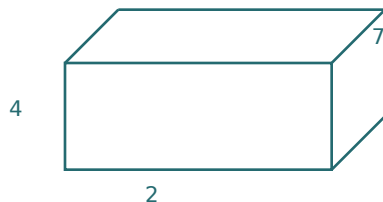
Explain your method.

Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

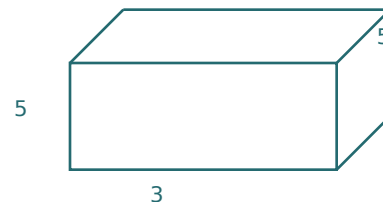
A storage prism for the music club measures 2 cm by 4 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 06

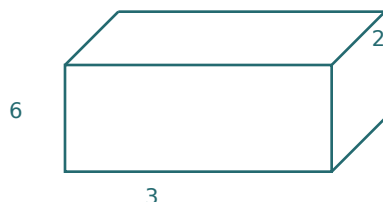
A storage prism for the library team measures 3 cm by 5 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 07

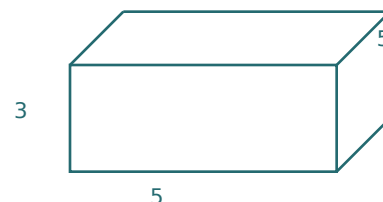
A storage prism for the reading group measures 3 cm by 6 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 08

A storage prism for the reading group measures 5 cm by 3 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



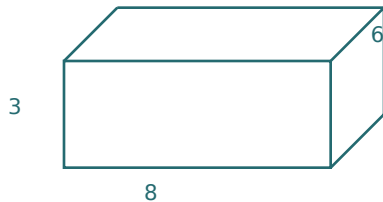
Explain your method.

Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

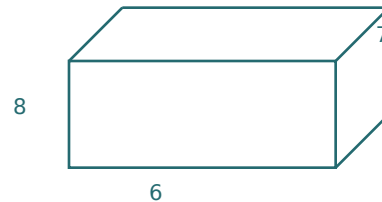
A storage prism for the music club measures 8 cm by 3 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 10

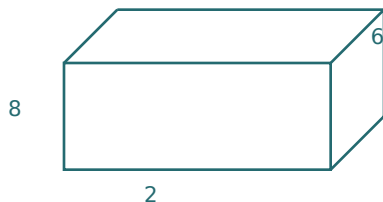
A storage prism for the school garden measures 6 cm by 8 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 11

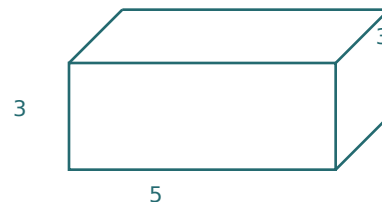
A storage prism for the art club measures 2 cm by 8 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 12

A storage prism for the design team measures 5 cm by 3 cm by 3 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



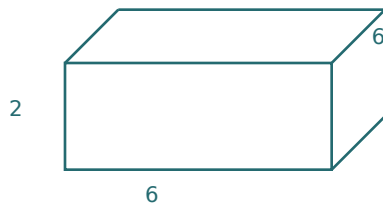
Explain your method.

Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

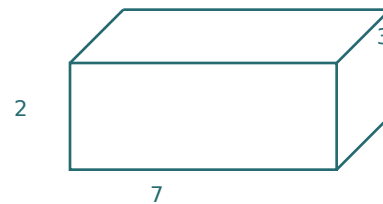
A storage prism for the community center measures 6 cm by 2 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 14

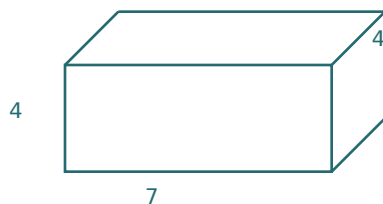
A storage prism for the school garden measures 7 cm by 2 cm by 3 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 15

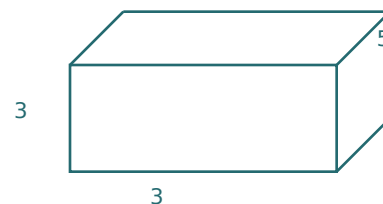
A storage prism for the library team measures 7 cm by 4 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

CARD 16

A storage prism for the reading group measures 3 cm by 3 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Explain your method.

Task-card response record

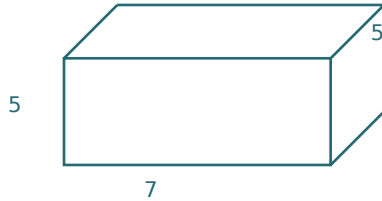
Use this page for final answers; attach drawings or full worked solutions on separate paper.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____

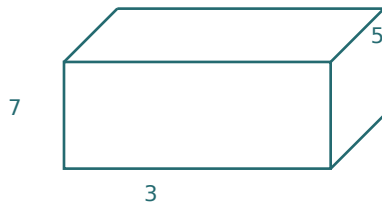
Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. A storage prism for the community center measures 7 cm by 5 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



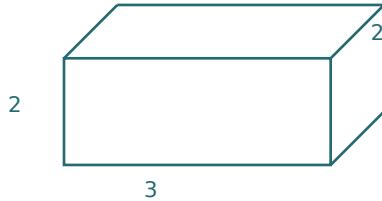
2. A storage prism for the library team measures 3 cm by 7 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



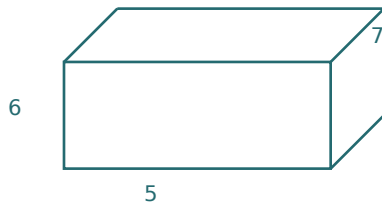
Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. A storage prism for the school garden measures 3 cm by 2 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



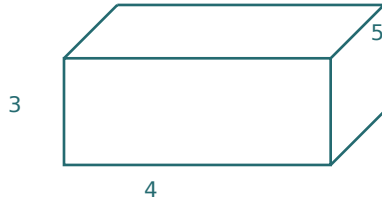
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4. A storage prism for the library team measures 5 cm by 6 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



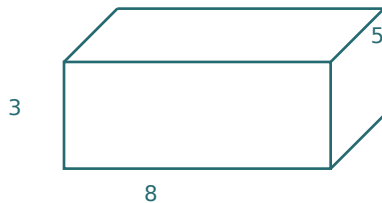
Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. A storage prism for the community center measures 4 cm by 3 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



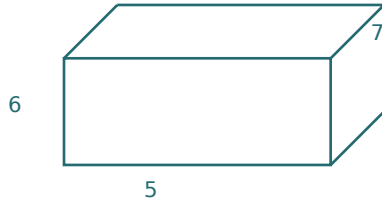
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6. A storage prism for the school garden measures 8 cm by 3 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



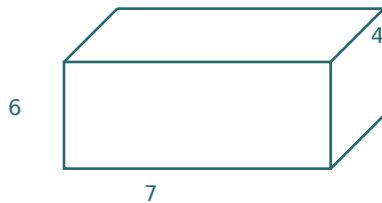
Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. A storage prism for the music club measures 5 cm by 6 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



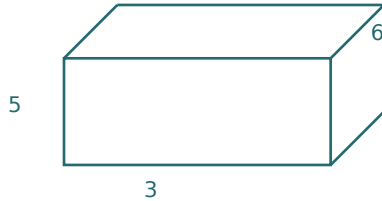
8. A storage prism for the reading group measures 7 cm by 6 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



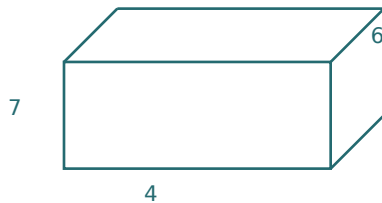
Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. A storage prism for the art club measures 3 cm by 5 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



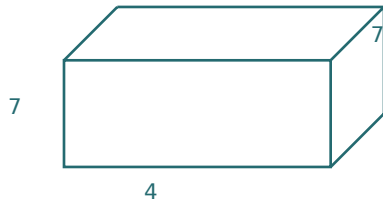
10. A storage prism for the community center measures 4 cm by 7 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



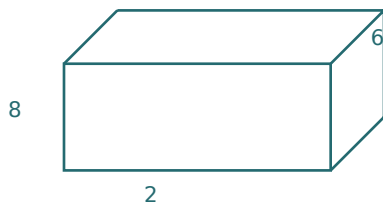
Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

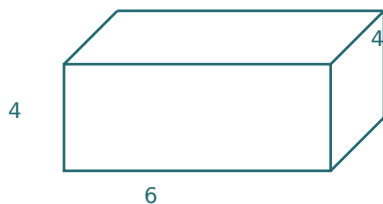
1. A storage prism for the community center measures 4 cm by 7 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



2. A storage prism for the library team measures 2 cm by 8 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



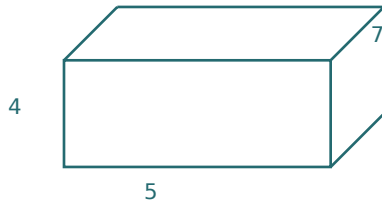
3. A storage prism for the design team measures 6 cm by 4 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



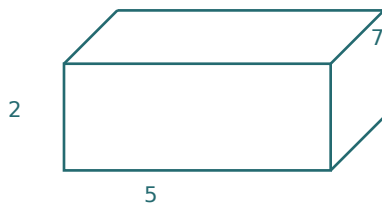
Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

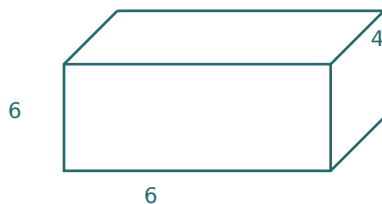
4. A storage prism for the school garden measures 5 cm by 4 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



5. A storage prism for the reading group measures 5 cm by 2 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



6. A storage prism for the science club measures 6 cm by 6 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. A storage prism for the music club measures 5 cm by 4 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

2. A storage prism for the art club measures 2 cm by 7 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

3. A storage prism for the design team measures 7 cm by 6 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

4. A storage prism for the design team measures 5 cm by 6 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Task-card worked solutions

1. 126 cm³

One layer has $6 \times 3 = 18$ cubes. 7 layers give $18 \times 7 = 126 \text{ cm}^3$.

2. 48 cm³

One layer has $3 \times 8 = 24$ cubes. 2 layers give $24 \times 2 = 48 \text{ cm}^3$.

3. 180 cm³

One layer has $6 \times 5 = 30$ cubes. 6 layers give $30 \times 6 = 180 \text{ cm}^3$.

4. 294 cm³

One layer has $7 \times 6 = 42$ cubes. 7 layers give $42 \times 7 = 294 \text{ cm}^3$.

5. 56 cm³

One layer has $2 \times 4 = 8$ cubes. 7 layers give $8 \times 7 = 56 \text{ cm}^3$.

6. 75 cm³

One layer has $3 \times 5 = 15$ cubes. 5 layers give $15 \times 5 = 75 \text{ cm}^3$.

Task-card worked solutions

7. 36 cm^3

One layer has $3 \times 6 = 18$ cubes. 2 layers give $18 \times 2 = 36 \text{ cm}^3$.

8. 75 cm^3

One layer has $5 \times 3 = 15$ cubes. 5 layers give $15 \times 5 = 75 \text{ cm}^3$.

9. 144 cm^3

One layer has $8 \times 3 = 24$ cubes. 6 layers give $24 \times 6 = 144 \text{ cm}^3$.

10. 336 cm^3

One layer has $6 \times 8 = 48$ cubes. 7 layers give $48 \times 7 = 336 \text{ cm}^3$.

11. 96 cm^3

One layer has $2 \times 8 = 16$ cubes. 6 layers give $16 \times 6 = 96 \text{ cm}^3$.

12. 45 cm^3

One layer has $5 \times 3 = 15$ cubes. 3 layers give $15 \times 3 = 45 \text{ cm}^3$.

Task-card worked solutions

13. 72 cm³

One layer has $6 \times 2 = 12$ cubes. 6 layers give $12 \times 6 = 72 \text{ cm}^3$.

14. 42 cm³

One layer has $7 \times 2 = 14$ cubes. 3 layers give $14 \times 3 = 42 \text{ cm}^3$.

15. 112 cm³

One layer has $7 \times 4 = 28$ cubes. 4 layers give $28 \times 4 = 112 \text{ cm}^3$.

16. 45 cm³

One layer has $3 \times 3 = 9$ cubes. 5 layers give $9 \times 5 = 45 \text{ cm}^3$.

Warm-up worked solutions

1. 175 cm^3

One layer has $7 \times 5 = 35$ cubes. 5 layers give $35 \times 5 = 175 \text{ cm}^3$.

2. 105 cm^3

One layer has $3 \times 7 = 21$ cubes. 5 layers give $21 \times 5 = 105 \text{ cm}^3$.

3. 12 cm^3

One layer has $3 \times 2 = 6$ cubes. 2 layers give $6 \times 2 = 12 \text{ cm}^3$.

4. 210 cm^3

One layer has $5 \times 6 = 30$ cubes. 7 layers give $30 \times 7 = 210 \text{ cm}^3$.

5. 60 cm^3

One layer has $4 \times 3 = 12$ cubes. 5 layers give $12 \times 5 = 60 \text{ cm}^3$.

6. 120 cm^3

One layer has $8 \times 3 = 24$ cubes. 5 layers give $24 \times 5 = 120 \text{ cm}^3$.

Warm-up worked solutions

7. 210 cm^3

One layer has $5 \times 6 = 30$ cubes. 7 layers give $30 \times 7 = 210 \text{ cm}^3$.

8. 168 cm^3

One layer has $7 \times 6 = 42$ cubes. 4 layers give $42 \times 4 = 168 \text{ cm}^3$.

9. 90 cm^3

One layer has $3 \times 5 = 15$ cubes. 6 layers give $15 \times 6 = 90 \text{ cm}^3$.

10. 168 cm^3

One layer has $4 \times 7 = 28$ cubes. 6 layers give $28 \times 6 = 168 \text{ cm}^3$.

Classwork worked solutions

1. 196 cm^3

One layer has $4 \times 7 = 28$ cubes. 7 layers give $28 \times 7 = 196 \text{ cm}^3$.

2. 96 cm^3

One layer has $2 \times 8 = 16$ cubes. 6 layers give $16 \times 6 = 96 \text{ cm}^3$.

3. 96 cm^3

One layer has $6 \times 4 = 24$ cubes. 4 layers give $24 \times 4 = 96 \text{ cm}^3$.

4. 140 cm^3

One layer has $5 \times 4 = 20$ cubes. 7 layers give $20 \times 7 = 140 \text{ cm}^3$.

5. 70 cm^3

One layer has $5 \times 2 = 10$ cubes. 7 layers give $10 \times 7 = 70 \text{ cm}^3$.

6. 144 cm^3

One layer has $6 \times 6 = 36$ cubes. 4 layers give $36 \times 4 = 144 \text{ cm}^3$.

Exit-ticket worked solutions

1. 120 cm³

One layer has $5 \times 4 = 20$ cubes. 6 layers give $20 \times 6 = 120 \text{ cm}^3$.

2. 98 cm³

One layer has $2 \times 7 = 14$ cubes. 7 layers give $14 \times 7 = 98 \text{ cm}^3$.

3. 168 cm³

One layer has $7 \times 6 = 42$ cubes. 4 layers give $42 \times 4 = 168 \text{ cm}^3$.

4. 180 cm³

One layer has $5 \times 6 = 30$ cubes. 6 layers give $30 \times 6 = 180 \text{ cm}^3$.

Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A storage prism for the September art display measures 5 cm by 6 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

2. A storage prism for the autumn reading fair measures 8 cm by 3 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

3. A storage prism for the September art display measures 2 cm by 7 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

4. A storage prism for the autumn reading fair measures 9 cm by 5 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A storage prism for the October book drive measures 4 cm by 3 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

6. A storage prism for the autumn reading fair measures 8 cm by 2 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

7. A storage prism for the October book drive measures 6 cm by 6 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

8. A storage prism for the leaf observation team measures 7 cm by 5 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Fall • Worked solutions

1. 210 cm³

One layer has $5 \times 6 = 30$ cubes. 7 layers give $30 \times 7 = 210$ cm³.

2. 144 cm³

One layer has $8 \times 3 = 24$ cubes. 6 layers give $24 \times 6 = 144$ cm³.

3. 56 cm³

One layer has $2 \times 7 = 14$ cubes. 4 layers give $14 \times 4 = 56$ cm³.

4. 90 cm³

One layer has $9 \times 5 = 45$ cubes. 2 layers give $45 \times 2 = 90$ cm³.

Fall • Worked solutions

5. 24 cm^3

One layer has $4 \times 3 = 12$ cubes. 2 layers give $12 \times 2 = 24 \text{ cm}^3$.

6. 80 cm^3

One layer has $8 \times 2 = 16$ cubes. 5 layers give $16 \times 5 = 80 \text{ cm}^3$.

7. 144 cm^3

One layer has $6 \times 6 = 36$ cubes. 4 layers give $36 \times 4 = 144 \text{ cm}^3$.

8. 175 cm^3

One layer has $7 \times 5 = 35$ cubes. 5 layers give $35 \times 5 = 175 \text{ cm}^3$.

Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A storage prism for the December art display measures 8 cm by 5 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

2. A storage prism for the December art display measures 3 cm by 5 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

3. A storage prism for the indoor garden project measures 6 cm by 7 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

4. A storage prism for the December art display measures 2 cm by 5 cm by 7 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A storage prism for the January puzzle club measures 3 cm by 4 cm by 3 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

6. A storage prism for the December art display measures 3 cm by 2 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

7. A storage prism for the indoor garden project measures 7 cm by 6 cm by 3 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

8. A storage prism for the winter supply drive measures 3 cm by 3 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Winter • Worked solutions

1. 280 cm^3

One layer has $8 \times 5 = 40$ cubes. 7 layers give $40 \times 7 = 280 \text{ cm}^3$.

2. 105 cm^3

One layer has $3 \times 5 = 15$ cubes. 7 layers give $15 \times 7 = 105 \text{ cm}^3$.

3. 294 cm^3

One layer has $6 \times 7 = 42$ cubes. 7 layers give $42 \times 7 = 294 \text{ cm}^3$.

4. 70 cm^3

One layer has $2 \times 5 = 10$ cubes. 7 layers give $10 \times 7 = 70 \text{ cm}^3$.

Winter • Worked solutions

5. 36 cm^3

One layer has $3 \times 4 = 12$ cubes. 3 layers give $12 \times 3 = 36 \text{ cm}^3$.

6. 36 cm^3

One layer has $3 \times 2 = 6$ cubes. 6 layers give $6 \times 6 = 36 \text{ cm}^3$.

7. 126 cm^3

One layer has $7 \times 6 = 42$ cubes. 3 layers give $42 \times 3 = 126 \text{ cm}^3$.

8. 36 cm^3

One layer has $3 \times 3 = 9$ cubes. 4 layers give $9 \times 4 = 36 \text{ cm}^3$.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A storage prism for the April science team measures 4 cm by 2 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

2. A storage prism for the March art display measures 6 cm by 4 cm by 3 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

3. A storage prism for the March art display measures 9 cm by 7 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

4. A storage prism for the spring book swap measures 8 cm by 8 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A storage prism for the March art display measures 7 cm by 3 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

6. A storage prism for the seedling club measures 5 cm by 8 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

7. A storage prism for the March art display measures 5 cm by 6 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

8. A storage prism for the seedling club measures 4 cm by 5 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Spring break • Worked solutions

1. 40 cm^3

One layer has $4 \times 2 = 8$ cubes. 5 layers give $8 \times 5 = 40 \text{ cm}^3$.

2. 72 cm^3

One layer has $6 \times 4 = 24$ cubes. 3 layers give $24 \times 3 = 72 \text{ cm}^3$.

3. 315 cm^3

One layer has $9 \times 7 = 63$ cubes. 5 layers give $63 \times 5 = 315 \text{ cm}^3$.

4. 128 cm^3

One layer has $8 \times 8 = 64$ cubes. 2 layers give $64 \times 2 = 128 \text{ cm}^3$.

Spring break • Worked solutions

5. 126 cm^3

One layer has $7 \times 3 = 21$ cubes. 6 layers give $21 \times 6 = 126 \text{ cm}^3$.

6. 80 cm^3

One layer has $5 \times 8 = 40$ cubes. 2 layers give $40 \times 2 = 80 \text{ cm}^3$.

7. 120 cm^3

One layer has $5 \times 6 = 30$ cubes. 4 layers give $30 \times 4 = 120 \text{ cm}^3$.

8. 100 cm^3

One layer has $4 \times 5 = 20$ cubes. 5 layers give $20 \times 5 = 100 \text{ cm}^3$.

Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A storage prism for the summer supply team measures 4 cm by 7 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

2. A storage prism for the summer supply team measures 9 cm by 6 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

3. A storage prism for the June art workshop measures 7 cm by 8 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

4. A storage prism for the summer library club measures 8 cm by 7 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A storage prism for the July puzzle camp measures 9 cm by 5 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

6. A storage prism for the July puzzle camp measures 4 cm by 2 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

7. A storage prism for the summer supply team measures 5 cm by 4 cm by 3 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

8. A storage prism for the July puzzle camp measures 9 cm by 3 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Summer • Worked solutions

1. 112 cm^3

One layer has $4 \times 7 = 28$ cubes. 4 layers give $28 \times 4 = 112 \text{ cm}^3$.

2. 324 cm^3

One layer has $9 \times 6 = 54$ cubes. 6 layers give $54 \times 6 = 324 \text{ cm}^3$.

3. 224 cm^3

One layer has $7 \times 8 = 56$ cubes. 4 layers give $56 \times 4 = 224 \text{ cm}^3$.

4. 224 cm^3

One layer has $8 \times 7 = 56$ cubes. 4 layers give $56 \times 4 = 224 \text{ cm}^3$.

Summer • Worked solutions

5. 225 cm³

One layer has $9 \times 5 = 45$ cubes. 5 layers give $45 \times 5 = 225$ cm³.

6. 40 cm³

One layer has $4 \times 2 = 8$ cubes. 5 layers give $8 \times 5 = 40$ cm³.

7. 60 cm³

One layer has $5 \times 4 = 20$ cubes. 3 layers give $20 \times 3 = 60$ cm³.

8. 108 cm³

One layer has $9 \times 3 = 27$ cubes. 4 layers give $27 \times 4 = 108$ cm³.

Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Connect layers of cubes with volume. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Count cubes in a base layer, then multiply by the number of layers. State volume in cubic units.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A storage prism for the library helpers measures 3 cm by 6 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

2. A storage prism for the library helpers measures 4 cm by 8 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

3. A storage prism for the library helpers measures 2 cm by 4 cm by 2 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

4. A storage prism for the classroom supply team measures 9 cm by 6 cm by 5 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Substitute lesson • Worked solutions

1. 72 cm^3

One layer has $3 \times 6 = 18$ cubes. 4 layers give $18 \times 4 = 72 \text{ cm}^3$.

2. 192 cm^3

One layer has $4 \times 8 = 32$ cubes. 6 layers give $32 \times 6 = 192 \text{ cm}^3$.

3. 16 cm^3

One layer has $2 \times 4 = 8$ cubes. 2 layers give $8 \times 2 = 16 \text{ cm}^3$.

4. 270 cm^3

One layer has $9 \times 6 = 54$ cubes. 5 layers give $54 \times 5 = 270 \text{ cm}^3$.

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Connect layers of cubes with volume. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Count cubes in a base layer, then multiply by the number of layers. State volume in cubic units.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A storage prism for the quiet design team measures 6 cm by 8 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

2. A storage prism for the classroom book group measures 3 cm by 5 cm by 6 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

3. A storage prism for the paper puzzle team measures 5 cm by 4 cm by 4 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.

4. A storage prism for the desk-supply team measures 8 cm by 8 cm by 3 cm. Find its volume. Explain the number of unit cubes in one layer and the number of layers.



Emergency substitute lesson • Worked solutions

1. 288 cm^3

One layer has $6 \times 8 = 48$ cubes. 6 layers give $48 \times 6 = 288 \text{ cm}^3$.

2. 90 cm^3

One layer has $3 \times 5 = 15$ cubes. 6 layers give $15 \times 6 = 90 \text{ cm}^3$.

3. 80 cm^3

One layer has $5 \times 4 = 20$ cubes. 4 layers give $20 \times 4 = 80 \text{ cm}^3$.

4. 192 cm^3

One layer has $8 \times 8 = 64$ cubes. 3 layers give $64 \times 3 = 192 \text{ cm}^3$.

A completed lesson-plan example

Level and focus

Grade 5; NC.5.MD.5; Connect layers of cubes with volume.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.5.MD.5 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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